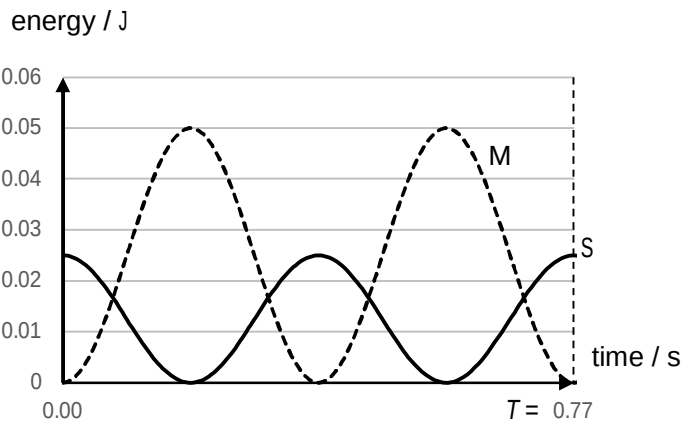


4	
(a)(i)	Maximum EPE stored in the 2 springs = $2 \times \frac{1}{2} kx^2 = 0.050 \text{ J}$
	When this is all transferred to KE $\frac{1}{2} m v_{\max}^2 = 0.050 \text{ J}$
	$v_{\max} = 0.41 \text{ m s}^{-1} \text{ (2sf)}$
(a)(ii)	$v_{\max} = \omega x_0$
	$T = 0.77 \text{ s (2sf)}$

(a)(iii)1. (a)(iii)2.	
	Correct M graph – $\sin^2$ and T or 0.77 shown
	Correct S graph – $\cos^2$ and $\frac{1}{2}$ energy of M
(b)	During an earthquake, as the building sways, energy is transferred to the mass-spring system of the damper. Therefore, amplitude of oscillation of the building decreases, preventing structural damage.
5	